

Tide retrospective: *linear response at any perturbation strength*

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Abstract

The susceptibility of the anharmonic Gibbs measure — the derivative of the perturbed mean $\langle u \rangle_h = G_1(h)/G_0(h)$ of $u = -(tx)$ — lifted from the single point $h = 0$ to an arbitrary base point $|h_0| < 1$:

$$\left. \frac{\partial}{\partial h} \right|_{h_0} \langle u \rangle_h = \frac{G_2(h_0)G_0(h_0) - G_1(h_0)^2}{G_0(h_0)^2}.$$

This is the fluctuation–dissipation (linear-response) identity at any perturbation strength, and the foundation for the third cumulant (the second derivative of the mean). New file `AnharmonicSusceptibilityGeneralH.lean`; builds clean first try, no `sorry/axiom/native_decide`.

1 Setting

Seventh tide of a single-day anharmonic arc on the laplace seabed. The previous tide¹ proved the susceptibility $\partial_h \langle u \rangle_h$ only at $h = 0$ (the connected variance κ_2). The third cumulant is the *second* derivative of the mean, so it needs the mean’s derivative as a function of h on a neighbourhood — which in turn needs the susceptibility at a general base point. This tide provides exactly that, at minimal risk: the general- h partition derivatives already exist (from the general- h step tide), so the only genuinely new pieces are positivity of the perturbed partition at $h_0 \neq 0$ and a pointwise quotient rule.

2 The thing formalised

For the anharmonic potential and $|h_0| < 1$, HasDerivAt of $h \mapsto G_1(h)/G_0(h)$ at h_0 with value $(G_2(h_0)G_0(h_0) - G_1(h_0)^2)/G_0(h_0)^2$; and $G_0(h) > 0$ for $|h| < 1$.

3 Proof strategy

Positivity. $G_0(h) = \int e^{-t(L+hx)}$ (after $-(tx)^0 = 1$) is a strictly-positive integrand with full support and is integrable at any $|h| < 1$ (the arc’s centre-integrability lemma at $n = 0$); so `integral_pos_iff_support_of_nonneg_ae` gives positivity — a line-for-line analogue of `anharmonic_partition_pos` with the $+hx$ carried through. **Susceptibility.** `HasDerivAt.div` on the two general- h partition derivatives — of G_1 (value $G_2(h_0)$) and of G_0 (value $G_1(h_0)$) at h_0 — with $G_0(h_0) \neq 0$.

¹2026-05-30-tide-anharmonic-second-cumulant.

4 Roadblocks and resolutions

None — it built on the first attempt. The general- h machinery from the earlier tides (centre-integrability, the general- h `HasDerivAt` step) is exactly the right interface; positivity reused the $h = 0$ template verbatim modulo the perturbation term.

5 What was Mathlib, what was new

Mathlib. `HasDerivAt.div`; the positivity-from-support lemma²; `Real.exp_ne_zero`; `Real.volume_univ`.

Seabed (reused). The general- h partition derivative and the centre-integrability lemma³, plus the $h = 0$ positivity template — all from this session’s arc.

New. General- h partition positivity and the general- h susceptibility. About 70 lines.

6 Lessons

- **Investing in the general- h step pays compounding dividends.** Because the partition derivatives were proved at an arbitrary centre two tides ago, the general- h susceptibility is just “the $h = 0$ proof with h_0 in place of 0” — no new analytic work, only the (reused) positivity and a quotient rule.
- **Positivity templates transfer across perturbations.** The $\int e^{-tL} > 0$ argument carries the $+hx$ term unchanged: the integrand is still everywhere positive with full support.

7 Follow-ups

- **Third cumulant κ_3 .** The second derivative of the mean: differentiate the general- h susceptibility’s mean function once more (the `iteratedDeriv` chaining pattern), then expand to $\langle u^3 \rangle_0 - 3\langle u^2 \rangle_0 \langle u \rangle_0 + 2\langle u \rangle_0^3$. Then κ_4 — the flow-equation identity (survey v4 #2).
- **Promote `amgm_t_abs_x` to lean-common.**

²`integral_pos_iff_support_of_nonneg_ae`.

³`weightedPartition.hasDerivAt` and `integrable.weightedPartition.integrand`.